Side-Channel Trojan Detection on FPGA

Physical-Region Test Generation and A/B Differential Results — s38417

This report summarises a differential side-channel experiment on the s38417 benchmark mapped to an FPGA. Two paired test sets were generated: **Dataset B** (MERS-s), engineered to maximise switching of the suspect rare logic, and **Dataset A**, a matched control engineered to minimise that switching while reproducing B’s background and input/output activity. The rare-node targeting used **physical FPGA regions** derived from Vivado placement (clustering radius = 6 slices). The results show Dataset A suppresses trigger switching by 96% relative to B while keeping all other activity matched — the condition that makes the differential measurement attributable to the Trojan trigger.

Parameter	Value
Benchmark	s38417 (Xilinx LUT/FDRE netlist)
Region type	Physical (FPGA placement-aware)
Clustering radius	6 slices (Euclidean, placement grid)
Dataset B	MERS-s (maximise rare/trigger switching)
Dataset A	Paired control (minimise trigger switching)
Vectors generated	900 per dataset (=> 899 transitions)
Trigger nodes measured	4 (Q_i_2__292, Q_i_2__363, Q_i_4__103, Q_i_6__5)

> **Headline result.** Dataset A reduces trigger-cone switching to **4% of Dataset B** (a 96% reduction), while background switching and input/output Hamming profiles stay matched within ~1.5%. This is the strongest A/B contrast obtained, and it was achieved with physical FPGA regions rather than logical netlist regions.

1. Results

1.1 Per-trigger P(at activating value)

For each trigger node this is the probability that the node sits at its *trigger-activating* value (shown as ==1 or ==0) across all test vectors. Dataset A is the quiet control, so a **lower** value in column A means better suppression — A keeps the trigger away from the value that would help fire the Trojan. The A/B ratio quantifies the gap.

Trigger node	A	B	A / B
Q_i_2__292_n_0 (==1)	0.444%	0.667%	0.67
Q_i_2__363_n_0 (==1)	0.222%	11.667%	0.02
Q_i_4__103_n_0 (==1)	2.333%	11.556%	0.20

Q_i_6__5_n_0 (==0)	0.667%	13.889%	0.05

Three of the four triggers are strongly suppressed: B drives them to the activating value 11–14% of the time, while A holds them at 0.2–2.3% (A/B ratios of 0.02–0.20). The first trigger shows a weak contrast (0.67) only because Dataset B itself rarely activated it (0.667%), so there was little switching to suppress.

1.2 Per-trigger toggle rate (switches per pair)

This is how often each trigger node *changes value* between consecutive vectors — its switching activity. This is the quantity a power side-channel actually senses: a toggling node draws dynamic power, a static one does not. Again **lower in A** is the goal.

Trigger node	A	B
Q_i_2__292_n_0	0.0089	0.0133
Q_i_2__363_n_0	0.0022	0.1913
Q_i_4__103_n_0	0.0011	0.0434
Q_i_6__5_n_0	0.0078	0.2347

A rate of 0.1913 means the node toggled on ~19% of the 899 transitions; 0.0022 means ~0.2%. Under Dataset B the trigger logic is electrically active; under Dataset A it is nearly frozen. This difference is the side-channel signal the method isolates.

1.3 Aggregate — the headline differential

Metric	Dataset A	Dataset B	Outcome
Trigger-cone switching / pair	0.0200	0.4828	**A = 4% of B**
'Any trigger switches' rate	2.00%	41.71%	**strongly quiet**
Pairs analysed (transitions)	899	899	**900 vectors - 1**

Summed across all four triggers, Dataset B produces ~0.48 trigger switches per transition; Dataset A produces ~0.02 — a **96% reduction**. On only 2% of transitions does any trigger move under A, versus 42% under B.

> **Why 900 vectors give 899 pairs.** Switching is measured *between* consecutive vectors, so a sequence of N vectors has N–1 transitions (like N fence posts having N–1 panels between them). The 900 vectors produce exactly 899 switching pairs. A toggle only exists between two vectors, never within one.

1.4 Matched-control verification

The suppression above is only meaningful if Dataset A reproduces Dataset B's *other* activity, so that a differential measurement cancels everything except the trigger. The table confirms this: background switching and input/output Hamming distance/weight are matched, while the rare cone is suppressed.

Metric	A: mean	B: mean	Verdict
RareConeSwitch	89.3	292.1	suppressed
BackgroundSwitch	3055.4	3103.9	matched (~1.5%)
Input HD	753.4	753.4	exact
Input HW	778.7	781.2	matched
Output HD	510.7	510.7	exact
Output HW	549.8	550.6	matched

RareConeSwitch here is measured over the whole physical-region cone (7,191 cone nodes, 35,810 background nodes), which is why it is a large count rather than near zero: with a cone that large, some node almost always switches, so the suppression appears as a 3.3× magnitude drop (292 → 89) rather than reaching exactly zero. The 4-trigger figures in section 1.3 are the focused view of the same effect.

> **Reading the two RareConeSwitch numbers.** Section 1.3 measures switching over just the 4 trigger nodes (0.02 vs 0.48). Section 1.4 measures it over the entire 7,191-node physical cone (89 vs 292). Different node sets, same conclusion: A is far quieter than B. The '== 0' counts being 0/900 for both simply reflects that a 7,191-node cone rarely has zero total switches.

2. Building physical FPGA regions

MERS normally excites *all* rare nodes, which is heavy and not aligned to where a Trojan can actually hide. An attacker can only build a trigger from rare signals that are physically close on the die, because the trigger gate must be routed to all of them. On an FPGA "physically close" means placed in nearby slices — and that is also what shares a power/EM signature, which is exactly what a side-channel measurement sees. So for SCA, regions should be defined by **placement proximity**, not by logical netlist structure.

> **Logical vs physical regions.** A gate-level netlist (.bench) contains no placement, so logical regions (shared fan-in cones) and the FPGA's physical regions are different partitions of the same circuit and generally do not match. For side-channel analysis the physical partition is the correct one, because power is spatial. This experiment therefore used physical regions built from Vivado placement.

2.1 Step-by-step procedure

Step 1 — Export placement from Vivado. After place-and-route (or with a routed checkpoint open), each leaf cell's slice location is written to a CSV. The provided Tcl script reads the cell's LOC / site and parses its SLICE_X#Y# column/row:

Step 1: placement export

```
# in the Vivado Tcl console, after place_design / open routed .dcp
```

```
source export_placement.tcl        ;# writes placement.csv (cell,x,y)

# placement.csv looks like:
cell,x,y
design_1/Q_i_2__292_n_0_reg,12,34
design_1/Q_i_2__363_n_0_reg,13,35
...
```

Step 2 — Detect rare nodes. Rare nodes are found by Monte-Carlo simulation (here using a large vector count for an accurate estimate). For s38417 this yields ~300 rare nodes after excluding synthetic LUT-artefact nodes.

Step 3 — Match rare nodes to coordinates. The rare-node names from the netlist are matched to Vivado cell names. Hierarchy prefixes (design_1/u_core/…) and register suffixes (_reg) are normalised away, so a name like Q_i_2__292_n_0 matches design_1/Q_i_2__292_n_0_reg. The match rate is reported; only located rare nodes are clustered.

Step 4 — Cluster by physical proximity. Each located rare node is a point (x, y) on the slice grid. Two rare nodes are linked if their Euclidean slice-distance is within the radius. In *compact* mode the algorithm grows size-capped clusters around the densest seeds:

```
Step 4: physical clustering (radius = 6 slices)
for every pair of placed rare nodes (i, j):
    dist = sqrt( (xi - xj)^2 + (yi - yj)^2 )      # slices
    if dist <= radius:  link i and j as neighbours

compact clustering:
    seed at the rare node with the most neighbours
    add nearest unused neighbours up to max_region_size
    remove them; repeat with the next densest seed

each region records: rare-node count, slice bounding-box (footprint),
                     density = rare nodes / footprint area, centroid
```

Step 5 — Rank and select the top-K regions. Regions are sorted by rare-node count (the densest physical pockets first). The top-K are the most probable Trojan locations; their rare nodes become the targeted set.

Step 6 — Generate the paired test sets on the selected region. MERS (Dataset B) excites only the rare nodes in the selected region(s), and Dataset A automatically reuses the same region selection as its suppression cone — so both test sets are aligned to the same physical pocket. The A/B differential in section 1 is then measured on that aligned target.

2.2 On the radius choice (6 slices)

The radius is a distance on the placement grid (in slices), not a physical length in mm/cm. A radius of 6 groups rare nodes lying within 6 slices of each other. For the s38417 placement used here (a roughly 60×134 slice area) radius 6 produces compact, trigger-sized pockets without merging unrelated logic; the resulting cone for the selected region(s) contained 7,191 nodes. A smaller radius fragments the pockets into singletons; a larger one risks merging physically separate clusters.

2.3 Physical regions formed (radius = 6)

Clustering the placed rare nodes at radius 6 produced **18 physical regions** (with no singleton rare nodes). The five densest are shown below. “Footprint” is the slice bounding-box area of the region; “density” is rare nodes per unit footprint — higher means a tighter physical pocket and a more probable Trojan hiding spot.

Region	Rare nodes	Footprint (slices)	Density (rare/area)
0	40	30	1.333
1	40	49	0.816
2	36	104	0.346
3	35	120	0.292
4	35	132	0.265

The top region packs 40 rare nodes into a 30-slice footprint (density 1.333) — an extremely dense pocket and the most likely place to hide a trigger. Targeting the **top-5 regions** selects **186 rare nodes** out of 302 total, a **38% reduction** in the set MERS must excite, focused on the densest physical clusters. Dataset A then reuses this same 186-node selection so both test sets target identical pockets.

> **Targeting summary.** 302 rare nodes detected → 18 physical regions formed at radius 6 → top-5 regions selected = 186 rare nodes (38% fewer). Both Dataset B (MERS) and Dataset A operate on this aligned 186-node target.

3. Test-set quality — side-channel sensitivity

Independently of the A/B differential, the generated Dataset B was scored for its raw Trojan-detection power using the **Side-Channel Sensitivity (SCS)** metric — specifically MaxRelSw, the maximum relative switching a test set induces in a Trojan's trigger logic. A higher SCS means the test set drives the hidden trigger harder, making it easier to see in a side-channel trace. Each MERS variant was compared against a random baseline on populations of 4-trigger and 8-trigger Trojans.

3.1 4-trigger Trojans

Method	SCS (MaxRelSw)	Size	vs Random
Random	0.000664	1,000	baseline
MERS	0.001073	900	1.62x
MERS-h	0.002150	900	3.24x
MERS-s	0.002224	900	3.35x

3.2 8-trigger Trojans

Method	SCS (MaxRelSw)	Size	vs Random
Random	0.000866	1,000	baseline
MERS	0.001542	900	1.78x
MERS-h	0.003519	900	4.06x
MERS-s	0.003587	900	4.14x

Across both Trojan populations the ordering is consistent: **MERS-s > MERS-h > MERS > Random**. MERS-s, the simulation-aware reordering used as Dataset B in this experiment, is the strongest — about **3.3×** the random baseline on 4-trigger Trojans and **4.1×** on 8-trigger Trojans, and it does so with fewer vectors (900 vs the random baseline's 1,000). The gain grows with trigger count, which is expected: harder-to-activate multi-input triggers benefit most from MERS's rare-node-aware ordering.

> **How this complements the A/B result.** SCS measures how strongly *Dataset B* excites a Trojan (detection power). The A/B differential in section 1 measures how cleanly *Dataset A* subtracts the background to isolate that excitation. Together: B is a high-sensitivity probe (3–4x random), and A is the matched quiet reference that makes the trigger's contribution stand out.

4. Conclusion

- Using **physical FPGA regions** (radius 6) to target the test generation, Dataset A suppressed trigger-cone switching to **4% of Dataset B (96% reduction)** — the strongest contrast achieved.
- Per-trigger, three of four trigger nodes were held at 0.2–2.3% activation versus 11–14% under B, with toggle rates an order of magnitude lower.
- The control remained **matched**: background switching within ~1.5% and input/output Hamming distance/weight essentially identical, so the differential is attributable to the trigger region.

- The 900-vector / 899-pair relationship is expected: switching is counted between consecutive vectors, giving N–1 transitions.
- Physical regions are the correct targeting basis for side-channel analysis because power is spatial; they are built from Vivado placement, independent of the logical netlist structure.
- At radius 6, 302 rare nodes formed 18 physical regions; the top-5 (186 rare nodes, 38% fewer) were targeted, with the densest packing 40 rare nodes into a 30-slice footprint.
- On raw detection power, **MERS-s** scored highest — ~3.3× the random baseline on 4-trigger Trojans and ~4.1× on 8-trigger Trojans — confirming Dataset B is a strong side-channel probe.